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Display device

Abstract

Disclosed is a display device including a controlling unit, a display driver electrically connected to the controlling unit, a sensor driver electrically connected to the controlling unit, a display element electrically connected to the display driver, and a sensor element electrically connected to the sensor driver. The display element comprises a first group of sub-pixels having a same color. The sensor element is sensing a biometric feature in a sensing time period, and the sensing time period comprises 3 to 120 frame times. All the first group of sub-pixels are turned on one time in each of the 3 to 120 frame times by the display driver.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. application Ser. No. 17/742,362, filed on May 11, 2022, which is a continuation application of U.S. application Ser. No. 16/416,263, filed on May 19, 2019. The contents of these applications are incorporated herein by reference.

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

(1) The present disclosure relates to a display device, and more particularly, to a display device having biometric sensors.

2. Description of the Prior Art

(2) The growing demand for information security and information privacy has driven use of biometric authentication on electronic devices such as smartphones, laptops, tablets, banking devices, and gaming consoles. A popular form of biometric authentication is fingerprint identification. Recently, fingerprint sensors have been adopted by various electronic devices so that the electronic devices can be unlocked by device owners via fingerprint authentication, protecting the electronic devices from unauthorized access.

(3) Conventionally, just for an example, a fingerprint sensor is provided separately from a display panel in a display device, so a screen-locked display device can be unlocked by simply touching the fingerprint sensor. Nevertheless, it is of great interest to display device manufacturers and users to combine the fingerprint sensor into the display panel, thereby increasing the screen-to-body ratio of the display device and offering a narrow-border or bezel-less design. However, it is difficult to add the fingerprint sensor into the display region without losing the resolution of the display device, and when the fingerprint sensor is disposed in the display region, the area of the fingerprint sensor will be too small, resulting in bad sensitivity.

SUMMARY OF THE DISCLOSURE

(4) According to an embodiment, a display device is disclosed and includes a controlling unit, a display driver electrically connected to the controlling unit, a sensor driver electrically connected to the controlling unit, a display element electrically connected to the display driver, and a sensor element electrically connected to the sensor driver. The display element comprises a first group of sub-pixels having a same color. The sensor element is sensing a biometric feature in a sensing time period, and the sensing time period comprises 3 to 120 frame times. All the first group of sub-pixels are turned on one time in each of the 3 to 120 frame times by the display driver, the sensor element is capable of receiving light emitted from the display element 3 to 120 times in the sensing time period to form a plurality of readout signals, and the sensor driver accumulates a portion of the readout signals and then receives the portion of the readout signals from the sensor element more than one time in the sensing time period, a last time of the more than one time is in an ending portion of the sensing time period.

(5) These and other objectives of the present disclosure will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** schematically illustrates a block diagram of a display device according to an embodiment of the present disclosure.

(2) FIG. **2** schematically illustrates a top view of a display device according to this embodiment of the present disclosure.

(3) FIG. **3** and FIG. **4** schematically illustrate sectional views of display elements and sensor

elements according to some embodiments of the present disclosure.
(4) FIG. **5** schematically illustrates a top view of an arrangement of the sub-pixels and the sensing units according to some embodiments of the present disclosure.
(5) FIG. **6** schematically illustrates a flowchart of a driving method of the display device for identifying the biometric feature according to this embodiment of the present disclosure.
(6) FIG. **7** schematically illustrates timing sequences of the display control signal, the sensor control signal, the display scanning signal, the sensor scanning signal and the readout signal.
(7) FIG. **8** schematically illustrates the sub-pixels turned on in different frame times according to some embodiments of the present disclosure.
(8) FIG. **9** schematically illustrates sub-pixels turned on in different frame times according to some embodiments of the present disclosure.
(9) FIG. **10** schematically illustrates the sub-pixels turned on in different frame times according to some embodiments of the present disclosure.
(10) FIG. **11** schematically illustrates the sub-pixels turned on in different frame times according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

(11) The present disclosure may be understood by reference to the following detailed description, taken in conjunction with the drawings as described below. It is noted that, for purposes of illustrative clarity and being easily understood by the readers, various drawings of this disclosure show a portion of the electronic device, and certain elements in various drawings may not be drawn to scale. In addition, the number and dimension of each element shown in drawings are only illustrative and are not intended to limit the scope of the present disclosure.
(12) Certain terms are used throughout the description and following claims to refer to particular elements. As one skilled in the art will understand, electronic equipment manufacturers may refer to an element by different names. This document does not intend to distinguish between elements that differ in name but not function. In the following description and in the claims, the terms “include”, “comprise” and “have” are used in an open-ended fashion, and thus should be interpreted to mean “include, but not limited to . . . ”.
(13) It will be understood that when an element or layer is referred to as being “disposed on” or “connected to” another element or layer, it can be directly on or directly connected to the other element or layer, or intervening elements or layers may be presented (indirectly). In contrast, when an element is referred to as being “directly on” or “directly connected to” another element or layer, there are no intervening elements or layers presented.
(14) In addition, in this specification, relative expressions may be used. For example, “lower”, “bottom”, “higher” or “top” are used to describe the position of one element relative to another.
(15) Although terms such as first, second, third, etc., maybe used to describe diverse constituent elements, such constituent elements are not limited by the terms. The terms are used only to discriminate a constituent element from other constituent elements in the specification. The claims may not use the same terms, but instead may use the terms first, second, third, etc. with respect to the order in which an element is claimed. Accordingly, in the following description, a first constituent element maybe a second constituent element in a claim.
(16) As used herein, the term “connected to” or “coupled to” that is used to designate a connection or coupling of one element to another element includes both a case that an element is “directly connected or coupled to” another element and a case that an element is “electronically connected or coupled to” another element via still another element.
(17) It should be noted that the technical features in different embodiments described in the following can be replaced, recombined, or mixed with one another to constitute another embodiment without departing from the spirit of the present disclosure.
(18) FIG. **1** schematically illustrates a block diagram of a display device according to an embodiment of the present disclosure. The display device **1** can both display image and identify

biometric feature in its display region. For example, the display device **1** may be a smart watch, a smartphone, a tablet or a display device having an in-display biometric sensor, but not limited thereto. In another embodiment, the display device **1** also could be used for tiling another display device to form a tiling electronic device. As shown in FIG. **1**, the display device **1** of this embodiment includes a controlling unit **10**, a display driver **12**, a sensor driver **14**, a display element **16**, and a sensor element **18**. The display driver **12** and the sensor driver **14** are electrically connected to the controlling unit **10** respectively, so the controlling unit **10** can respectively provide a display control signal DC to the display driver **12** and provide a sensor control signal SC to the sensor driver **14** for separately controlling on/off state of the display driver **12** and on/off state of the sensor driver **14**. The controlling unit **10** may be for example an image processor, a digital signal processor, a central processing unit (CPU), a microprocessor or other suitable device. Furthermore, the display element **16** is electrically connected to and driven by the display driver **12**, and the sensor element **18** is electrically connected to and driven by the sensor driver **14**. After receiving the display control signal DC, the display driver **12** can transmit a display driving signal DDS to the display element **16** to turn on the display element **16**; similarly, after receiving the sensor control signal SC, the sensor driver **14** can transmit a sensor driving signal SDS to the sensor element **18** to turn on the sensor element **18**. The display element **16** is used for displaying image on screen or generating light for biometric feature identification; for example, the display element **16** may include sub-pixels of self-emissive display panel, such as an organic light-emitting diode (OLED) display panel, or an inorganic LED (such as quantum-dot LED, Mini-LED, or Micro-LED) display panel, or sub-pixels of non-self-emissive display panel, such as a liquid crystal display panel. The sensor element **18** is used for receiving reflected light from the biometric feature, so as to capture information (such as an image) of the biometric feature. The sensor element **18** maybe for example a fingerprint sensor or other kind of image sensor, but not limited thereto.

(19) In this embodiment, the display device **1** may optionally further include a processing unit **24**, and the controlling unit **10** is electrically connected to the display driver **12** and the sensor driver **14** through the processing unit **24**. The function of the processing unit **24** may be the same as or similar to the function of the controlling unit **10**, but not limited thereto. For example, the processing unit **24** can drive or operate the display driver **12** and drive the sensor driver **14**, and then read signals from the sensor element **18** out. In such case, the controlling unit **10** may not only be used to drive the display driver **12**, the sensor driver **14**, the display element **16** and the sensor element **18**, but also other devices. Thus, when the area of the display element **16** and the sensor element **18** is large, the data needed to process is huge, which may reduce the signal processing speed of the controlling unit **10**. By means of the processing unit **24** designed to process signals in advance, the load of the controlling unit **10** can be shared with the processing unit **24**. The signal processing speed of the display device **1** may be enhanced accordingly. In such situation, the display control signal DC and the sensor control signal SC may be provided from the processing unit **24**. In some embodiment, the display driver **12**, the sensor driver **14** and the processing unit **24** may be integrated into a single integrated circuit **25** that may be bonded on an array substrate of the display device **1**, and the display element **16** and the sensor element **18** may be electrically connected to the single integrated circuit **25** through wires, but not limited thereto. In some embodiments, the display device **1** may not include the processing unit **24**.

(20) The display driver **12** may include a gate driver **121** and a data driver **122** for driving the display element **16**. The gate driver **121** may provide scanning signals for turning on the display element **16**, and the data driver **122** may provide display signals to the display element **16**, such that the display element **16** can display the desired image. For example, the gate driver **121** may include shift register unit, logic unit, level shift unit and/or digital buffer unit, and the data driver **122** may include shift register unit, level shift unit, digital to analog converter (or multiplexer) and/or analog buffer unit, but not limited thereto. The sensor driver **14** may include another gate

driver **141** and a readout driver **142** for signal readout of the sensor element **18**. The gate driver **141** may provide scanning signals for turning on the sensor element **18**, and the readout driver **142** may receive readout signals from the sensor element **18**, such that the signals detected by the sensor element **18** can be read out. In some embodiments, the gate driver **121**, the data driver **122**, the gate driver **141** and/or the readout driver **142** may be formed in the array substrate of the display device **1** by thin-film transistor process, but not limited thereto. In some embodiments, the gate driver **121**, data driver **122**, the gate driver **141** and/or the readout driver **142** may be formed in one or more integrated circuit and be bonded on the array substrate or a flexible printed circuit substrate that adheres to the array substrate.

(21) In this embodiment, the display driver **12** may optionally further include a timing control unit **123** electrically connected to the gate driver **121** and the data driver **122** to provide timing signals to the gate driver **121** and the data driver **122** respectively, and the gate driver **121** and the data driver **122** are electrically connected to the controlling unit **10** or the processing unit **24** through the timing control unit **123**. In some embodiments, the timing control unit **123** may be disposed outside the display driver **12** but still be electrically connected between the gate driver **121** and the controlling unit **10** or the processing unit **24** and between the data driver **122** and the controlling unit **10** or the processing unit **24**.

(22) In this embodiment, the sensor driver **14** may optionally further include another timing control unit **143** electrically connected to the gate driver **141** and the readout driver **142** to provide timing signals to the gate driver **141** and the readout driver **142** respectively, and the gate driver **141** and the readout driver **142** are electrically connected to the controlling unit **10** or the processing unit **24** through the another timing control unit **143**. In some embodiments, the another timing control unit **143** may be disposed outside the sensor driver **14** but still be electrically connected between the gate driver **141** and the controlling unit **10** or the processing unit **24** and between the readout driver **142** and the controlling unit **10** or the processing unit **24**.

(23) In some embodiments, the timing control unit **123** and the another timing control unit **143** may be integrated into one timing control unit outside the display driver **12** and the sensor driver **14**, and the integrated timing control unit may electrically connect the display driver **12** and the sensor driver **14** to the controlling unit **10** or the processing unit **24** respectively, thereby reducing the number of the timing control units. Since the display driver **12** and the sensor driver **14** are controlled by the same one timing control unit, the possibility of timing disorder or timing error can be reduced. Also, when the processing unit **24**, the integrated one timing control unit, the display driver **12** and the sensor driver **14** are integrated into one integrated circuit, the size of the integrated circuit can be decreased, and the border width of the display device **1** for disposing the integrated circuit can be narrowed. In some embodiments, the integrated one timing controlling unit and the processing unit **24** maybe integrated into a single integrated circuit while the display driver **12** and the sensor driver **14** are not integrated into the single integrated circuit.

(24) In some embodiments, the display device **1** may further include a touch sensing element (not shown in figures) for sensing the position of the biometric feature. The touch sensing element may be electrically connected to the controlling unit **10** or the processing unit **24**, such that when the touch sensing element senses the touching of the biometric feature on the display device **1**, the controlling unit **10** or the processing unit **24** may define the touching region of the biometric feature as a biometric feature identifying region for further detecting the biometric feature.

(25) FIG. **2** schematically illustrates a top view of a display device according to this embodiment of the present disclosure. The display device **1** has a display region **1**D for displaying image and a peripheral region **1**P that doesn't display image. The peripheral region **1**P is disposed on at least one side of the display region **1**D. For example, the peripheral region **1**P surrounds the display region **1**D. A width of the peripheral region **1**P may be also referred to as the border width of the display decide **1**. The display element **16** and the sensor element **18** are disposed in the display region **1**D, and the display driver **12**, the sensor driver **14** and the processing unit **24** shown in FIG. **1** are

disposed in the peripheral region **1**P. The regions and the numbers of the gate driver **121** and the data driver **122** of display driver **12** and the regions and numbers of the gate driver **141** and the readout driver **142** of the sensor driver **14** shown in FIG. **2** are for illustration and not limited thereto. In some embodiments, the number of the gate driver **121** and/or the number of the data driver **122** may be one or more. In some embodiments, the number of the gate driver **141** and/or the number of the readout driver **142** may be one or more.

(26) Specifically, the display device **1** may further include a plurality of display scan lines **26**, a plurality of display data lines **28**, a plurality of sensor scan lines **30**, and a plurality of sensor signal lines **32** formed on a substrate Sub. The display element **16** includes a plurality of sub-pixels **161** and a plurality of transistors **162** formed on the substrate Sub, in which each sub-pixel **161** is electrically connected to a corresponding transistor **162**, and a gate and a source of each transistor **162** are electrically connected to the gate driver **121** and the data driver **122** respectively through a corresponding display scan line **26** and a corresponding display data line **28**, such that the on/off state of each transistor **162** can be controlled by display scanning signals DS**1**-DSN from the gate driver **121**, and the display of each sub-pixel **161** can be controlled by display data signals DD**1**-DDN from the data driver **122**. In one embodiment, the display element **16** may include at least one transistor **162** corresponding to one of the sub-pixels **161**, for example, if the display element **16** includes at least two transistors **162** corresponding to one of the sub-pixels **161**, wherein the at least two transistors **162** could be of a same type or different types, but not limited thereto. In an example of the transistor of the display element **16** electrically connected to the corresponding sub-pixel **161**, the number of the transistor corresponding to one of the sub-pixels may be 1, 3, 4 or 7. In another example of the display element **16** including plural transistors **162** corresponding to one sub-pixel **161**, the plural transistors **162** may include a switch transistor and/or a driving transistor, but not limited thereto. In some embodiments, the transistor **162** “corresponding to” one of the sub-pixels as used herein represents the transistor **162** is electrically connected to the one of the sub-pixels **161**.

(27) Similarly, the sensor element **18** includes a plurality of sensing units **181** and a plurality of transistors **182** formed on the substrate Sub, in which each sensing unit **181** is electrically connected to a corresponding transistor **182**, and a gate and a source of each transistor **182** are electrically connected to the gate driver **141** and the driver **142** respectively through a corresponding sensor scan line **30** and a corresponding sensor signal line **32**, such that the on/off state of each transistor **182** can be controlled by sensor scanning signals SS**1**-SSN from the gate driver **141**, and the readout driver **142** may receive readout signals RS**1**-RSN from each sensing unit **181**. In addition, the sensor element **18** may include at least one transistor **182** corresponding to one of the sensing units **181**. In another example of the transistor of the sensor element **18** electrically connected to the corresponding sensing unit **181**, the number of the transistor may be 2 or 3. In another example of the sensor element **18** including plural transistors **182** corresponding to one sensing unit **181**, the plural transistors **182** may include a readout transistor, a reset transistor and/or an amplifier transistor but not limited thereto. The transistor **182** “corresponding to” one of the sensing units **181** as used herein represents the transistor **182** is electrically connected to the one of the sensing units **181**. In some embodiments, the transistor **162** and/or the transistor **182** may be a thin-film transistor. In some embodiments, type of the transistor **162** and/or the transistor **182** may be an amorphous thin-film transistor, a low-temperature polysilicon thin-film transistor, a metal-oxide thin-film transistor, but it is not limited thereto.

(28) In this embodiment, each sensing unit **181** is used to detect light from the corresponding sub-pixel **161**. For example, the sensing units **181** may include a plurality of sensing units **181***a*, a plurality of sensing units **181***b*, and a plurality of sensing units **181***c*, the sub-pixels **161** may include a plurality of first sub-pixel **161***a* for emitting light with a first color, a plurality of second sub-pixel **161***b* for emitting light with a second color, and a plurality of third sub-pixel **161***c* for emitting light with a third color, and each sensing unit **181***a*, each sensing unit **181***b* and each

sensing unit **181***c* correspond to one first sub-pixel **161***a*, one second sub-pixel **161***b* and one third sub-pixel **161***c* respectively. In other words, each sensing unit **181***a* can detect the light having the first color, each sensing unit **181***b* can detect the light having the second color, and each sensing unit **181***c* can detect the light having the third color, in which the first color, the second color and the third color are different from one another, but not limited thereto. For example, the first color, the second color and the third color may be respectively red, green and blue, but not limited thereto. For example, the sensing unit **181** may include a corresponding color filter disposed thereon for allowing one color passing through and filtering light with different colors so as to avoid light interference. In some embodiments, each sensing unit **181** may for example be an optical (such as photo diode) detector, a capacitive detector, a radio frequency (RF) detector, a thermal, piezoresistive detector, an ultrasonic detector, or a piezoelectric detector, but not limited thereto.

(29) FIG. **3** and FIG. **4** schematically illustrate sectional views of display devices according to some embodiments of the present disclosure. In some embodiments shown in FIG. **3**, the display device **1**A may include self-emissive display panel, and each of the sub-pixels **161** of the display element **16**A may include a light-emitting unit LU. Specifically, the display device **1**A may include a display layer **34**A formed on the substrate Sub and a cover substrate **36** covering and protecting the display layer **34**A, in which the display layer **34**A may include the light-emitting units LU of the display element **16**A and the sensor element **18** including a plurality of sensing units **181**, and the sensor element **18** may be formed under the light-emitting units LU, but not limited thereto. For example, the sensor element **18** is disposed between the light-emitting units LU and the substrate Sub, i.e. the light-emitting units LU and the sensor element **18** are formed in the same display layer **34**A. In a vertical direction VD (top view), the sensing units **181** may be disposed in the gaps of the sub-pixels **161**, such that the light from the sub-pixel **161** can be reflected to the corresponding sensing unit **181** by the biometric feature **38**, as an arrow shown in FIG. **3**. The vertical direction VD is defined as being substantially perpendicular to a horizontal direction HD. The horizontal direction HD represents the horizontal direction of the substrate Sub that is flattened horizontally. In some embodiments, the light-emitting unit LU may include organic light-emitting diode (OLED), quantum-dot LED (QLED), Mini-LED, Micro-LED or other type light-emitting device. In some embodiments, the sensing units **181** may be formed under the substrate Sub, so that the sub-pixels **161** and the sensing units **181** are formed on two sides of the substrate Sub (not shown in FIG. **3**).

(30) In some embodiments shown in FIG. **4**, the display device **1**B may include non-self-emissive display panel. For example, the non-self-emissive display panel is a liquid crystal display panel, and the display layer **34**B of the display device **1**B may include a liquid crystal layer LC, the display element **16**B and the sensor element **18** between the substrate Sub and a counter substrate **40**, in which the liquid crystal layer LC is disposed between the display element **16**B and the sensor element **18**. Specifically, the display element **16**B is disposed between the liquid crystal layer LC and the substrate Sub, the sensor element **18** is disposed between the liquid crystal layer LC and a counter substrate **40**, and the sensing unit **181** is disposed above the corresponding transistor (e.g. transistor **162**) of the display element **16**B. Also, the display device **1**B may further include a backlight module BLU disposed under the substrate Sub and used for generating a suitable light (such as white light). The sub-pixel **161** of some embodiments may include a liquid crystal controller **1611** for controlling the rotation of the liquid crystal molecules corresponding to the region of the sub-pixel **161**. For example, the liquid crystal controller **1611** may be a pixel electrode or other components, but not limited thereto. Thus, light from backlight module BLU can penetrate through the sub-pixel **161** and be reflected by the biometric feature to the corresponding sensing unit **181**, as an arrow shown in FIG. **4**. For example, the counter substrate **40** may include color filters (not shown) for allowing light having different colors to be emitted. In some embodiments, in order to generate light with uniform brightness, the color filters with different

colors may have different thicknesses. In some embodiments, the display device **1**B may optionally include the cover substrate **36** disposed on the counter substrate **40**. In some embodiments, the sensor element **18** may be disposed between the counter substrate **40** and the cover substrate **36**. In one embodiment, the cover substrate **36** can have a touch function, but not limited thereto.
(31) FIG. **5** schematically illustrates a top view of an arrangement of the sub-pixels and the sensing units according to some embodiments of the present disclosure. In some embodiments, the arrangement of the sub-pixels **161** may be pentile matrix, and each sensing unit **181** is disposed between adjacent two of sub-pixels **161**. For example, in an odd row, the first sub-pixels **161***a* having first color and the third sub-pixels **161***c* having third color are alternately arranged along the row direction, and the sensing units **181** and the sub-pixels **161** (such as **161***a* and **161***c*) in the odd row are alternately arranged along the row direction. In an even row, the second sub-pixels **161***b* having second color are arranged along the row direction, and the sub-pixels **161** (such as **161***b*) and the sensing units **181** in the even row are alternately arranged along the row direction. Also, in an odd column, the second sub-pixels **161***b* and the sensing units **181** are alternately arranged along the column direction; and in an even column, the first sub-pixels **161***a* and the third sub-pixels **161***c* are alternately arranged along the column direction, and the sub-pixels **161** (such as **161***a* and **161***c*) and the sensing units **181** in the odd column are alternately arranged along the column direction.
(32) In some embodiments, the first sub-pixel **161***a*, the second sub-pixel **161***b* and the third sub-pixel **161***c* may have different areas in a top view of the display device **1**. For example, since the first color, the second color and the third color are red, green and blue respectively, the area of the third sub-pixel **161***c* may be greater than the area of the first sub-pixel **161***a*, and the area of the first sub-pixel **161***a* may be greater than the area of the second sub-pixel **161***b*. In some embodiments, the sensing units **181** may have different areas in a top view of the display device **1** based on the area difference between the sub-pixels **161**. For example, the sensing units **181** arranged in the same row as the first sub-pixels **161***a* and the third sub-pixels **161***c* (such as odd row) may have greater area or less area than the sensing units **181** arranged in the same row as the second sub-pixels **161***b* (such as even row). In some embodiments, one of the sensing units **181** between the adjacent first sub-pixel **161***a* and the adjacent third sub-pixel **161***c* may be closer to the adjacent first sub-pixel **161***a* having the first color than the adjacent third sub-pixel **161***c* having the third color. In other words, since third sub-pixel **161***c* emits blue light, which is scattered easier and worse than the red light, to improve the signal to noise ratio (S/N) of the sensor element **18**, a distance W**1** between the sensing unit **181** and the adjacent first sub-pixel **161***a* is less than a distance W**2** between the sensing unit **181** and the adjacent third sub-pixel **161***c*. The distance W**1** is the shortest distance between the sensing unit **181** and the first sub-pixel **161***a* adjacent to each other in a row direction RD, and the distance W**2** is the shortest distance between the sensing unit **181** and the third sub-pixel **161***c* adjacent to each other in the row direction RD. In some embodiments, any two of the colors of the first sub-pixel **161***a*, the second sub-pixel **161***b* and the third sub-pixel **161***c* may be exchanged or may be other colors. In some embodiments, arrangement of the first sub-pixel **161***a*, the second sub-pixel **161***b* and the third sub-pixel **161***c* may be other kind of arrangement.
(33) The following description further details a driving method of the display device of the present disclosure. FIG. **6** schematically illustrates a flowchart of a driving method of the display device for identifying the biometric feature according to this embodiment of the present disclosure, and FIG. **7** schematically illustrates timing sequences of the display control signal, the sensor control signal, the display scanning signal, the sensor scanning signal and the readout signal. The driving method of the display device includes the following steps and is detailed accompanying with FIG. **7** as well as FIG. **1** and FIG. **2**. In step S**102**, the display device **1** starts a sensing time period SP. In the sensing time period SP, the display device **1** starts to detect the biometric feature. For example, when the sensor element **18** detects the touching of the biometric feature on the display device **1**, the controlling unit **10** or the processing unit **24** may start the sensing time period SP for detecting

the biometric feature. The condition for starting the sensing time period SP of the present disclosure is not limited thereto.
(34) In step S**104**, the display control signal DC and the sensor control signal SC (an enable signal) are respectively provided to the display driver **12** and the sensor driver **14** by the control unit **10** or the processing unit **24**, such that the sensor driver **14** can be activated by the enable signal from the control unit **10** or the processing unit **24**. The duration of the display control signal DC overlaps the duration of the sensor control signal SC, such that the display element **16** may generate light for biometric feature identification and the sensor element **18** can detect the reflected light from the biometric feature. For example, the display control signal DC and the sensor control signal SC may be simultaneously provided by the controlling unit **10** or the processing unit **24** and be ended at the same time, such that the display driver **12** and the sensor driver **14** can be synchronized by the controlling unit **10** or the processing unit **24**. In some embodiments, the start time ST**1** of the display control signal DC may be earlier than the start time ST**2** of sensor control signal SC. As long as the biometric feature can be correctly identified, the start time ST**1** and the end time ET**1** of the display control signal DC and the start time ST**2** and the end time ET**2** of the sensor control signal SC can be altered.
(35) In step S**106**, after the display driver **12** receives the display control signal DC, a display driving signal DDS is transmitted to a display element **16** by the display driver **12**, and the display element **16** is refreshed a first predetermined number of times when the sensing element **18** is sensing the biometric feature in the sensing time period SP. In this embodiment, the first predetermined number of times may range from 3 to 120 in the sensing time period SP. In other words, the display driving signal DDS may be repeatedly transmitted to the display element **16** ranged from 3 to 120 times, such that the sub-pixels **161** may generate light pulse 3 to 120 times. It is noted that since the display element **16** can generate light pulse 3 to 120 times in the sensing time period SP, the sensor element **18** can receive light 3 to 120 times, thereby accumulating enough electric charges and improving the S/N ratio and the sensitivity of the sensing units. For this reason, the sensing units **181** can be disposed in the display region **1**D without reducing the area of each sub-pixel **161** or reducing the resolution of the display device **1**, i.e. the sensor element **18** can be integrated with the display element **16** into the display region **1**D of the display device **1**, thereby increasing screen-to-body ratio while providing biometric authentication. Also, even the area of the sensing unit **181** is small, through the increased refreshing times, the sensing units **181** still can have good sensitivity.
(36) Specifically, the display driving signal DDS may include a plurality of display scanning signals DS**1**-DSN transmitted to the display scan line **26** respectively and a plurality of display data signals DD**1**-DDN transmitted to the display data lines **28** respectively. In one frame time FT of the sensing time period SP, the display scanning signals DS**1**-DSN may be sequentially transmitted one time, and the display data signals DD**1**-DDN may be transmitted according to the display scanning signals DS**1**-DSN, such that the sub-pixels **161** can emit the required light pulse one time while receiving the display data signals DD**1**-DDN. When the display element **16** is refreshed 3 to 120 times in the sensing time period SP, the display scanning signals DS**1**-DSN may be sequentially transmitted to the display element **16** ranged from 3 to 120 times in 3 to 120 frame times FT. In some embodiments, adjacent two of the display scanning signals DS**1**-DSN may overlap each other or not in one frame time FT. In some embodiment, the first predetermined number of times may be 6 to 72 times. In order to have short unlocking time, the first predetermined number of times may be decreased based on the frame time FT. For example, when the refreshing rate(frequency) of the display device **1** is 60 Hz (i.e. the frame time FT is 0.0167 seconds, the first predetermined number of times may be 6, 18 or 30. When the refreshing rate(frequency) of the display device **1** is 120 Hz (i.e. the frame time FT is 0.0083 seconds less than 0.0167 seconds), the first predetermined number of times may be increased to be 12, 36 or 60.
(37) In step S**108**, after the sensor driver **14** receives the sensor control signal SC, a sensor driving

signal SDS is transmitted to the sensor element **18** by the sensor driver **14**, and the sensor element **18** is refreshed a second predetermined number of times in the sensing time period SP. In this embodiment, as shown in FIG. **7**, the second predetermined number of times may be one. Specifically, the sensor driving signal SDS may include a plurality of sensor scanning signals SS**1**-SSN and a plurality of readout signals RS**1**-RSN. When the sensor element **18** is driven once, the duration of the sensor scanning signals SS**1**-SSN may overlap all the frame times FT of the display element **16**. The readout signals RS**1**-RSN may be received by the readout driver **142** near the end time of the sensor scanning signals SS**1**-SSN. The duration of the readout signals RS**1**-RSN may overlap at least one frame time FT or not. In some embodiments, the second predetermined number of times may be plural, so the electric charges converted by the sensing units **181** can be read out plural times in one sensing time period SP by the readout driver **142**, and the readout signals RS**1**-RSN received by the readout driver **142** more than one time in one sensing time period SP can be accumulated by the sensor driver **14** to increase the S/N ratio. In some embodiments, the sensor element **18** of the display device **1** may perform more than one time to detect the biometric feature in more than one sensing time period SP.

(38) FIG. **8** schematically illustrates the sub-pixels turned on in different frame times according to some embodiments of the present disclosure. For clarity, the arrangement of the sub-pixels mentioned in the following embodiments takes pentile matrix for an example, and sensing units are not shown, but not limited thereto. In some embodiments, all the sub-pixels **161** having the same color are turned on repeatedly, and the remaining sub-pixels having different colors are turned off in one sensing time period, so as to display one frame in each frame time. For example, all the second sub-pixels **161***b* having the second color are driven and turned on repeatedly to sequentially generate the frames (e.g. a first frame F**1**A, a second frame F**2**A, etc.) while the first sub-pixels **161***a* and the third sub-pixels **161***c* are turned off, so as to generate light having the same second color in each frame time. Thus, the sensing units **181** corresponding to the second sub-pixels **161***b* may detect the light having the second color without being interfered by other color light, thereby increasing the sensitivity of the sensing units **181**. For example, a circle shown in FIG. **8** may represent a region of reflected light from each second sub-pixel **161***b*. In some embodiments, all the sub-pixels **161** turned on in each frame FA may also be all the first sub-pixels **161***a* or all the third sub-pixels **161***c*. In one embodiment, a region shape of a reflected light from one sub-pixel **161** may be a circle, a polygon or a free-shape, but not limited thereto. In some embodiments, an area of the region is not limited to the drawings in the figures and also could be adjusted to a suitable size, but not limited thereto.

(39) FIG. **9** schematically illustrates sub-pixels turned on in different frame times according to some embodiments of the present disclosure. In some embodiments, a first group G**1** of the sub-pixels **161** and a second group G**2** of the sub-pixels **161** are turned on sequentially and alternately, and the remaining sub-pixels **161** are turned off in the sensing time period SP, in which the sub-pixels **161** in first group G**1** are different from the sub-pixels **161** in the second group G**2**. Specifically, the first group G**1** of the sub-pixels **161** is turned on to display one frame FIB in one frame time, and then the second group G**2** of the sub-pixels **161** is turned on to display another frame F**2**B in another frame time, in which the first group G**1** of the sub-pixels **161** and the second group G**2** of the sub-pixels **161** may have the same color. As an example, the second sub-pixels **161***b* may be the sub-pixels **161** turned on in the sensing time period. The second sub-pixels **161***b* in the first group G**1** are not adjacent sub-pixels **161**, so the distance between the adjacent second sub-pixels **161***b* in the first group G**1** can be increased, thereby reducing the interference of reflected light from different second sub-pixels **161***b*, as the circles shown in FIG. **9**. Similarly, the second sub-pixels **161***b* in the second group G**2** may prevent the interference of different second sub-pixels **161***b*. For example, one of the second sub-pixels **161***b* in the first group G**1** maybe disposed between adjacent two of the second sub-pixels **161***b* in the second group G**2**. In other words, the second sub-pixels **161***b* in the first group G**1** and the second sub-pixels **161***b* in the

second group G**2** are respectively arranged in staggered pattern, and in each even row, each second sub-pixel **161***b* in the first group G**1** and each second sub-pixel **161***b* in the second group G**2** are alternately arranged.

(40) FIG. **10** schematically illustrates the sub-pixels turned on in different frame times according to some embodiments of the present disclosure. In some embodiments, as compared with FIG. **9**, the first group G**1** of the sub-pixels **161** and the second group G**2** of the sub-pixels **161** turned on sequentially may have different colors. Specifically, the first group G**1** of the sub-pixels **161** having one color is turned on to display one frame FIC of one frame time, and the second group G**2** of the sub-pixels **161** having another color is then turned on to display another frame F**2**C of another subsequent frame time while the remaining sub-pixels **161** are turned off. In some embodiment, after the frame F**2**C, a third group G**3** of the sub-pixel **161** of the same color as the first group G**1** maybe turned on to display another frame F**3**C of another subsequent frame time, and then the second group G**2** of the sub-pixels **161** is turned on to display another frame F**4**C of another subsequent frame time, so the first group G**1** of the sub-pixels **161**, the second group G**2** of the sub-pixels **161**, the third group G**3** of the sub-pixels **161** and the second group G**2** of the sub-pixels **161** are turned on alternately. As an example, a part of the second sub-pixels **161***b* may be the first group G**1** of the sub-pixels **161**, other part of the second sub-pixels **161***b* may be the third group G**3** of the sub-pixels **161**, and all the first sub-pixels **161***a* may be the second group G**2** of the sub-pixels **161**. The second sub-pixels **161***b* in the first group G**1** may be the same as the first group G**1** shown in FIG. **9**, and the second sub-pixels **161***b* in the third group G**3** may be the same as the second group G**2** in FIG. **9**, so the patterns of the second sub-pixels in first group G**1** and third group G**3** will not be detailed redundantly. Through alternately turning on the first group G**1**, the second group G**2**, the third group G**3** and the second group G**2**, the frames F**1**C, F**2**C, F**3**C, F**4**C can be alternately generated. It is noted that through alternately turning on the groups of the sub-pixels with different colors, more sensing units can be used to detect the biometric feature, and interference of reflected light from different sub-pixels **161***b* of different colors may still be reduced.

(41) In some embodiments, the first group G**1** of the sub-pixels **161** may be all the sub-pixels **161** having the same color (e.g. the second sub-pixels **161***b*), the second group G**2** of the sub-pixels **161** may be all the sub-pixels **161** having another color (e.g. the first sub-pixels **161***a*), and the first group G**1** and the second group G**2** may be alternately turned on to display the frames F**1**C, F**2**C alternately.

(42) FIG. **11** schematically illustrates the sub-pixels turned on in different frame times according to some embodiments of the present disclosure. In some embodiments, the first group G**1** of the sub-pixels **161** and the second group G**2** of the sub-pixels **161** that have different colors maybe turned on at a same time. Specifically, the first group G**1** of the sub-pixels **161** having one color and the second group G**2** of the sub-pixels **161** having another color are turned on to display one frame FID of one frame time while the remaining sub-pixels **161** are turned off. In some embodiments, after the frame F**1**D, a third group G**3** of the sub-pixel **161** having the same color as the first group G**1** and the second group G**2** may be turned on to display another frame F**2**D of another subsequent frame time, so the frames F**1**D, F**2**D may be displayed alternately. As an example, apart of the second sub-pixels **161***b* may be the first group G**1** of the sub-pixels **161**, and other part of the second sub-pixels **161***b* may be the third group G**3** of the sub-pixels **161**, and all the first sub-pixels **161***a* may be the second group G**2** of the sub-pixels **161**. The second sub-pixels **161***b* in the first group G**1** may be the same as the first group G**1** shown in FIG. **9**, and the second sub-pixels **161***b* in the third group G**3** may be the same as the second group G**2** in FIG. **9**, so the patterns of the second sub-pixels in first group G**1** and third group G**3** will not be detailed redundantly. For example, the first sub-pixels **161***a* have one of red and green, and the second sub-pixels **161***b* have the other one of red and green.

(43) In some embodiments, after the frame F**1**D, the first group G**1** of the sub-pixels **161** having

one color and the second group G**2** of the sub-pixels **161** having another color may be repeatedly turned on to display another frame F**2**D in next frame time. In some embodiments, all the second sub-pixels **161***b* may be the first group G**1** of the sub-pixels **161**.

(44) As mentioned above, the display device can generate light pulse 3 to 120 times in the sensing time period, so the sensor element can receive light 3 to 120 times to accumulate enough signals and improve the S/N ratio and the sensitivity of the sensing units. For this reason, the sensor element can be integrated with the display element into the display region of the display device without reducing the area of each sub-pixel or reducing the resolution of the display device, thereby increasing screen-to-body ratio while providing biometric authentication.

(45) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the disclosure. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A display device, comprising: a controlling unit; a display driver electrically connected to the controlling unit; a sensor driver electrically connected to the controlling unit; a display element electrically connected to the display driver; and a sensor element electrically connected to the sensor driver; wherein the display element comprises a plurality of sub-pixels, and the plurality of sub-pixels are composed of a first group of sub-pixels having a first color and a second group of sub-pixels having a second color different from the first color, wherein the sensor element comprises a plurality of sensing units each respectively corresponding to one of the plurality of sub-pixels, the sensor element is configured to sense a biometric feature in a sensing time period, the sensing time period comprises 3 to 120 frame times, wherein during each of the 3 to 120 frame times, at least one of the plurality of sub-pixels emits light, and at least one of the corresponding sensing units receives the light, wherein during the sensing time period, the first group of sub-pixels is turned on one time in a first frame time of the 3 to 120 frame times by the display driver, while the second group of sub-pixels is turned off during the first frame time of the 3 to 120 frame times by the display driver, the second group of sub-pixels is turned on one time in a second frame time of the 3 to 120 frame times by the display driver, the second frame time is sequential with the first frame time in the sensing time period, while the first group of sub-pixels is turned off during the second frame time of the 3 to 120 frame times by the display driver, the sensor element is capable of receiving light emitted from the display element 3 to 120 times in the sensing time period to form a plurality of readout signals, and the sensor driver accumulates a portion of the plurality of readout signals and then receives the portion of the plurality of readout signals from the sensor element more than one time in the sensing time period, a last time of the more than one time is in an ending portion of the sensing time period.

2. The display device according to claim 1, wherein the display element further comprises a third group of sub-pixels having the first color, the third group of sub-pixels is turned off during the first frame time and the second frame time, and the third group of sub-pixels is turned on one time in a third frame time of the 3 to 120 frame times by the display driver, the third frame time is sequential with the second frame time in the sensing time period, while the first group of sub-pixels and the second group of sub-pixels are turned off during the third frame time of the 3 to 120 frame times by the display driver.

3. The display device according to claim 1, wherein the display element further comprises a plurality of other sub-pixels, and the plurality of other sub-pixels are turned off during the first frame time and the second frame time.

4. The display device according to claim 1, wherein the first group of sub-pixels comprise a first sub-pixel, the second group of sub-pixels comprise a second sub-pixel, one of the plurality of

sensing units is disposed between the first sub-pixel and the second sub-pixel, and a distance between the one of the plurality of sensing units and the first sub-pixel is less than a distance between the one of the plurality of sensing units and the second sub-pixel.

5. The display device according to claim 1, further comprising a liquid crystal layer disposed between the display element and the sensor element.

6. **1**. A display device, comprising: a controlling unit; a display driver electrically connected to the controlling unit; a sensor driver electrically connected to the controlling unit; a display element electrically connected to the display driver; and a sensor element electrically connected to the sensor driver, wherein the sensor element comprises a plurality of sensing units, and two of the sensing units have different areas; and a plurality of display scan lines, wherein the display element comprises a first group of sub-pixels having a first color and a second group of sub-pixels having a second color different from the first color, the first group of sub-pixels and the second group of sub-pixels have different areas, the sensor element is configured to sense a biometric feature in a sensing time period, the sensing time period comprises 3 to 120 frame times, a plurality of display scanning signals DS**1**-DSN are sequentially transmitted to the plurality of display scan lines one time in each of the 3 to 120 frame times by the display driver, respectively, the sensor element is capable of receiving light emitted from the display element 3 to 120 times in the sensing time period to form a plurality of readout signals, the sensor driver accumulates a portion of the plurality of readout signals and then receives the portion of the plurality of readout signals from the sensor element more than one time in the sensing time period, and a last time of the more than one time is in an ending portion of the sensing time period.